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Electronic device for controlling attributes of one-handed mode screen on basis of grip shape and method for controlling same

Abstract

Disclosed are an electronic device for controlling attributes of one-handed mode screen on the basis of a grip shape and a method for controlling same. An electronic device according to an embodiment of the present document may be configured to display, on a flexible display, a one-handed mode screen which is reduced and displayed according to a predetermined ratio so as to correspond to a first screen, detect, while the one-handed mode screen is displayed, that a first housing is moved, and determine, on the basis of the detection of the movement of the first housing, the direction of expansion of the one-handed mode screen depending on whether a grip of a portion of a user's body on the top portion of the first housing is sensed by a first grip sensor, among grip sensors, which is disposed on the edge of the first housing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation application, claiming priority under § 365 (c), of International Application No. PCT/KR2023/004214, filed on Mar. 29, 2023, which is based on and claims the benefit of a Korean patent application number 10-2022-0043869, filed on Apr. 8, 2022, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2022-0103172, filed on Aug. 18, 2022, in the Korean Intellectual Property Office, the disclosures of each of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(1) The disclosure relates to an electronic device for controlling, based on a grip type, properties of a one-handed mode screen, and a method for controlling the same.

2. Description of Related Art

(2) Various services and additional functions provided through an electronic device, for example, a portable electronic device such as a smartphone, are gradually increasing. In order to increase the utility value of such electronic device and satisfy the needs of various users, communication service providers or electronic device manufacturers are competitively developing electronic devices which provide various functions and differentiate the communication service providers or the electronic device manufacturers from other companies. Accordingly, various functions provided through the electronic device are also increasingly advanced.

SUMMARY

(3) A user of an electronic device (e.g., a smartphone) may use one hand to control (e.g., touch) a screen in which the size of a screen (e.g., a home screen) being displayed on the electronic device is reduced by a designated ratio. Herein, for convenience of description, such a function or operation may be referred to as “one-handed manipulation mode” or “one-handed keyboard mode”. When the “one-handed manipulation mode” or the “one-handed keyboard mode” is used, the screen having a size reduced according to the designated ratio may be displayed at a position in which the one-handed manipulation mode was last used. Therefore, if the one-handed mode is executed in the opposite direction to the position of the hand gripping the electronic device (e.g., when the screen reduced in size according to the designated ratio is displayed), the user may experience the inconvenience of having to re-position the screen. In addition, when a housing slides (e.g., the electronic device is switched from a closed state to an opened state) while a one-handed manipulation mode is used in an electronic device in which a flexible display is extended in the

vertical direction, a one-handed manipulation mode screen (simply, a “one-handed mode screen”) that is currently displayed on the electronic device may be extended and displayed toward an upper end by the sliding. In this case, a portion extended to the upper end may be displayed in a position in which it is difficult for the user to control the portion in the one-handed manipulation mode. Therefore, in order to continuously use the one-handed manipulation mode, a user setting for adjusting the display position of the one-handed mode screen may be required, which may make the user feel uncomfortable.

(4) An embodiment herein provides an electronic device wherein when the electronic device is switched from a closed state to an opened state, properties of a one-handed mode screen may be controlled (e.g., the extension direction of the one-handed mode screen may be determined) according to the state of a user's grip on the electronic device, thereby reducing the user's inconvenience in the use of a one-handed manipulation mode.

(5) An embodiment herein provides a method for controlling an electronic device, wherein when the electronic device is switched from a closed state to an opened state, properties of a one-handed mode screen may be controlled (e.g., the extension direction of the one-handed mode screen may be determined) according to the state of a user's grip on the electronic device, thereby reducing the user's inconvenience in the use of a one-handed manipulation mode.

(6) However, the problems to be solved in the disclosure are not limited to the above-mentioned problems, and may be variously expanded without departing from the spirit and scope of the disclosure.

(7) An electronic device according to an embodiment herein may include a first housing, a second housing disposed to be movable with respect to the first housing, at least one processor disposed in the second housing, and grip sensors disposed at edges of the first housing and the second housing, respectively, and a flexible display at least partially mounted on the surface of the second housing and at least partially exposed to the outside of the electronic device, wherein a portion of the flexible display is inserted into or withdrawn from the first housing by sliding of the first housing, and the at least one processor is configured to display, on the flexible display, a one-handed mode screen that is reduced and displayed according to a predetermined ratio corresponding to a first screen, detect that the second housing is moved while the one-handed mode screen is displayed, and determine, based on the detection of the movement of the second housing, the extension direction of the one-handed mode screen according to whether a first grip sensor disposed at the edge of the first housing among the grip sensors senses that a user's body portion grips an upper end of the first housing.

(8) A method for controlling an electronic device, according to an embodiment herein, may include displaying, on the flexible display, a one-handed mode screen that is reduced according to a predetermined ratio corresponding to a first screen and displayed, detecting that the second housing is moved while the one-handed mode screen is displayed, and determining, based on the detection of the movement of the second housing, the extension direction of the one-handed mode screen according to whether a first grip sensor disposed at the edge of the first housing among the grip sensors senses that a user's body portion grips an upper end of the first housing.

(9) An embodiment herein provides an electronic device wherein when the electronic device is switched from a closed state to an opened state, properties of a one-handed mode screen may be controlled (e.g., the extension direction of the one-handed mode screen may be determined) according to the state of a user's grip on the electronic device, thereby reducing the user's inconvenience in the use of a one-handed manipulation mode.

(10) Effects according to various embodiments herein are not limited to the above-described effects, and it is apparent to those skilled in the art that various effects are inherent in the disclosure

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a block diagram of an electronic device in a network environment according to various embodiments.
- (2) FIG. 2 illustrates a state in which a second display area of a display according to various embodiments herein is accommodated in a housing.
- (3) FIG. 3 illustrates a state in which the second display area of the display according to various embodiments herein is exposed to the outside of the housing.
- (4) FIG. 4 is an exploded perspective view of an electronic device according to various embodiments herein.
- (5) FIG. 5A is a cross-sectional view taken along line A-A' in FIG. 2 according to various embodiments herein.
- (6) FIG. 5B is a cross-sectional view taken along line B-B' in FIG. 3 according to various embodiments herein.
- (7) FIG. 6 is an illustrative flowchart illustrating functions or operations in which a property of a one-handed mode screen is changed depending on a grip type in which a user grips an electronic device when the state of an electronic device according to an embodiment herein is changed.
- (8) FIG. 7A is an exemplary view illustrating a screen for configuring a one-handed manipulation mode according to an embodiment herein.
- (9) FIG. 7B is an exemplary view illustrating a function or operation of entering a one-handed manipulation mode according to a user gesture according to an embodiment herein.
- (10) FIG. 7C is an exemplary view illustrating a function or operation of entering a one-handed manipulation mode according to a user's selection input for a visual object according to an embodiment herein.
- (11) FIGS. 8A and 8B are exemplary views illustrating the art related to embodiments herein, in which a one-hand mode screen is displayed regardless of the type of a user's grip.
- (12) FIG. 9A is an exemplary view illustrating a grip sensing area according to an embodiment herein.
- (13) FIGS. 9B and 9C are exemplary views illustrating a grip sensing area of an electronic device according to another embodiment herein.
- (14) FIG. 10A is an exemplary view illustrating a function or operation in which the extension direction of a one-handed mode screen is determined based on a first grip type when an electronic device according to an embodiment herein is switched from a closed state to an opened state.
- (15) FIG. 10B is an exemplary view illustrating a function or operation in which the extension direction of a one-handed mode screen is determined based on a second grip type when an electronic device according to an embodiment herein is switched from a closed state to an opened state.
- (16) FIG. 11A is an exemplary view illustrating a function or operation in which the extension direction of a one-handed mode screen is determined based on a first grip type when an electronic device according to an embodiment herein is switched from an opened state to a closed state.
- (17) FIG. 11B is an exemplary view illustrating a function or operation in which the extension direction of a one-handed mode screen is determined based on a second grip type when an electronic device according to an embodiment herein is switched from an opened state to a closed state.
- (18) FIG. 12 is an exemplary flowchart illustrating a function or operation in which a one-handed mode screen is moved and displayed according to the type of a user's grip when an electronic device according to an embodiment herein is in a closed state.
- (19) FIGS. 13A, 13B, and 13C are exemplary views illustrating the function or operation exemplarily described in FIG. 12 in terms of a user interface.

DETAILED DESCRIPTION

(20) FIG. 1 is a block diagram illustrating an electronic device **101** in a network environment **100** according to various embodiments. Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

(21) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

(22) The auxiliary processor **123** may control, for example, at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active (e.g., executing an application) state. According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence model is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-

network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

(23) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

(24) The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

(25) The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

(26) The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

(27) The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

(28) The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or an external electronic device (e.g., an electronic device **102** (e.g., a speaker or a headphone)) directly or wirelessly coupled with the electronic device **101**.

(29) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(30) The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

(31) A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

(32) The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

(33) The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

(34) The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

(35) The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

(36) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device **104** via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify or authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

(37) The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(38) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board

(PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

(39) According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mm Wave antenna module may include a printed circuit board, an RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(40) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(41) According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the external electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

(42) FIG. 2 illustrates a state in which a second display area of a display according to various embodiments herein is accommodated in a housing. FIG. 3 illustrates a state in which the second display area of the display according to various embodiments herein is exposed to the outside of the housing.

(43) FIGS. 2 and 3 show a structure in which a display **203** (e.g., a flexible display or a rollable display) is extended in the longitudinal direction (e.g., the +Y direction) when viewed from the front of an electronic device **101**. However, the extension direction of the display **203** is not limited to one direction (e.g., the +Y direction). For example, the extension direction of the display **203** can be designed/changed such that the display **203** is extendable in the upward direction (+Y direction),

the rightward direction (e.g., the +X direction), the leftward direction (e.g., the -X direction), and/or the downward direction (e.g., the -Y direction)

(44) The state illustrated in FIG. 2 may be referred to as a closed state of the electronic device **101** or a housing **210** and a slide-in state of the display **203**.

(45) The state illustrated in FIG. 3 may be referred to as an opened state of the electronic device **101** or the housing **210**, and a slide-out state of the display **203**.

(46) Referring to FIGS. 2 and 3, the electronic device **101** may include the housing **210**. The housing **210** may include a first housing **201** and a second housing **202** movably disposed relative to the first housing **201**. In an embodiment, it may be construed that the electronic device **101** has a structure in which the first housing **201** is disposed to be slidable with respect to the second housing **202**. According to one embodiment, the second housing **202** may be arranged to reciprocate a predetermined distance in an illustrated direction with respect to the first housing **201**, for example, a direction indicated by arrow  custom character.

(47) According to various embodiments, the second housing **202** may be referred to as a slide unit or a slide housing, and may be movable relative to the first housing **201**. According to one embodiment, the second housing **202** may accommodate various electrical and electronic components such as a circuit board or a battery.

(48) According to one embodiment, a motor, a speaker, a sim socket, and/or a sub circuit board electrically connected to a main circuit board (e.g., a second circuit board **249** in FIG. 4) may be disposed in the first housing **201**. The second housing **202** may accommodate the main circuit board (e.g., the first circuit board **248** in FIG. 4) on which electrical components such as an application processor (AP) and a communication processor (CP) are mounted.

(49) According to various embodiments, the first housing **201** may include a first cover member **211** (e.g., a main case). The first cover member **211** may include a (1-1)th side wall **211a**, a (1-2)th side wall **211b** extending from the (1-1)th side wall **211a**, and a (1-3)th side wall **211c** extending from the (1-1)th side wall **211a** and substantially parallel to the (1-2)th side wall **211b**. According to an embodiment, the (1-2)th side wall **211b** and the (1-3)th side wall **211c** may be formed substantially perpendicular to the (1-1)th side wall **211a**.

(50) According to various embodiments, the (1-1)th side wall **211a**, the (1-2)th side wall **211b**, and the (1-3)th side wall **211c** of the first cover member **211** may be formed in a shape in which one side (e.g., a front face) is open so as to accommodate (or surround) at least a portion of the second housing **202**. For example, at least a portion of the second housing **202** may be surrounded by the first housing **201**, and may slide in a direction parallel to a first surface (e.g., the first surface F1 in FIG. 4), for example, in the direction of arrow  custom character, while being guided by the first housing **201**. According to an embodiment, the (1-1)th side wall **211a**, the (1-2)th side wall **211b**, and/or the (1-3)th side wall **211c** of the first cover member **211** may be integrally formed. According to another embodiment, the (1-1)th side wall **211a**, the (1-2)th side wall **211b**, and/or the (1-3)th side wall **211c** of the first cover member **211** may be formed as separate structures to be coupled or assembled.

(51) According to various embodiments, the first cover member **211** may be formed to surround at least a portion of the display **203**. For example, at least a portion of the display **203** may be formed to be surrounded by the (1-1)th side wall **211a**, the (1-2)th side wall **211b**, and/or the (1-3)th side of the first cover member **211**.

(52) According to various embodiments, the second housing **202** may include a second cover member **221** (e.g., a slide plate). The second cover member **221** may have a plate shape, and may include a first surface (e.g., the first surface F1 in FIG. 4) for supporting internal components. For example, the second cover member **221** may support at least a portion (e.g., a first display area A1) of the display **203**. According to an embodiment, the second cover member **221** may be referred to as a front cover.

(53) According to one embodiment, the second cover member **221** may include a (2-1)th side wall

221a, a (2-2)th side wall **221b** extending from the (2-1)th side wall **221a**, and a (2-3)th side wall **221c** extending from the (2-1)th side wall **221a** and substantially parallel to the (2-2)th side wall **221b**. According to an embodiment, the (2-2)th side wall **221b** and the (2-3)th side wall **221c** may be formed substantially perpendicular to the (2-1)th side wall **221a**.

(54) According to various embodiments, the second housing **202** may form an opened state and a closed state of the housing **210** by moving in a first direction (e.g., direction  custom character parallel to the (1-2)th side wall **211b** or the (1-3)th side wall **211c**. In the closed state, the second housing **202** may move to be positioned at a first distance from the (1-1)th side wall **211a**, and in the opened state, the second housing **202** may move to be positioned at a second distance greater than at a first distance from the (1-1)th side wall **211a**. In an embodiment, in the closed state, the first housing **201** may surround a portion of the (2-1)th side wall **221a**.

(55) According to various embodiments, the electronic device **101** may include the display **203**, a key input device **245**, a connector hole **243**, audio modules **247a** and **247b**, or camera modules **249a** and **249b**. According to an embodiment, the electronic device **101** may further include an indicator (e.g., an LED device) or various sensor modules.

(56) According to various embodiments, the display **203** may include a first display area **A1** and a second display area **A2** configured to be exposed to the outside of the electronic device **101** based on sliding of the second housing **202**. According to an embodiment, the first display area **A1** may be disposed on the second housing **202**. For example, the first display area **A1** may be disposed on the second cover member **221** of the second housing **202**. According to an embodiment, the second display area **A2** may extend from the first display area **A1**, and may be accommodated in the first housing **201** (e.g., in a slide-in state) or visually exposed to the outside of the electronic device **101** (e.g., in a slide-out state) depending on sliding of the second housing **202** relative to the first housing **201**.

(57) According to various embodiments, the second display area **A2** may be moved while being substantially guided by one area (e.g., a curved surface **213a** in FIG. 4) of the first housing **201**, and may be accommodated in the inner space of the first housing **201** or exposed to the outside of the electronic device **101**. According to an embodiment, the second display area **A2** may be moved based on the sliding of the second housing **202** in the first direction (e.g., the direction indicated by arrow (1)). For example, while the second housing **202** slides, a portion of the second display area **A2** may be deformed into a curved shape at a position corresponding to the curved surface **213a** of the first housing **201**.

(58) According to various embodiments, when viewed from above the second cover member **221** (e.g., the front cover), if the housing **210** is switched from a closed state to an opened state (e.g., if the second housing **202** slides to extend from the first housing **201**), the second display area **A2** may be gradually exposed to the outside of the first housing **201** to form a substantially flat surface with the first display area **A1**. According to an embodiment, the display **203** may be coupled to or disposed adjacent to a touch sensing circuit, a pressure sensor capable of measuring the intensity (pressure) of a touch, and/or a digitizer for detecting a magnetic field-type stylus pen. According to an embodiment, regardless of the closed state or the opened state of the housing **210**, the exposed portion of the second display area **A2** may be positioned on a portion (e.g., curved surface **213a** in FIG. 4) of the first housing. In addition, a portion of the second display area **A2** at a position corresponding to the curved surface **213a** may maintain a curved shape.

(59) According to various embodiments, the key input device **245** may be positioned in one area of the first housing **201**. Depending on the appearance and usage state, the electronic device **101** may be designed such that the illustrated key input device **245** is omitted or an additional key input device (additional key input devices) is (are) included. According to an embodiment, the electronic device **101** may include an unillustrated key input device, for example, a home key button or a touch pad disposed around the home key button. According to an embodiment, at least a portion of the key input device **245** may be disposed on the (1-1)th side wall **211a**, the (1-2)th side wall **211b**,

and/or the (1-3) side wall **211c** of the first housing **201**.

(60) According to various embodiments, the connector hole **243** may be omitted according to embodiments, and may accommodate a connector (e.g., a USB connector) for transmitting/receiving power and/or data to and from an external electronic device. According to an embodiment (not shown), the electronic device **101** may include multiple connector holes **243**, and some of the multiple connector holes **243** may function as connector holes for transmitting and receiving audio signals to and from an external electronic device. In the illustrated embodiment, the connector hole **243** is positioned in the second housing **202**. However, the disclosure is not limited thereto, and the connector hole **243** or an unillustrated connector hole may be positioned in the first housing **201**.

(61) According to various embodiments, the audio modules **247a** and **247b** may include at least one speaker hole **247a**, or at least one microphone hole **247b**. One of the speaker holes **247a** may be provided as a receiver hole for a voice call, and the other may be provided as an external speaker hole. The electronic device **101** may include a microphone for acquiring a sound, and the microphone may acquire a sound outside the electronic device **101** through the microphone hole **247b**. According to an embodiment, the electronic device **101** may include multiple microphones to detect the direction of sound. According to an embodiment, the electronic device **101** may include an audio module in which the speaker hole **247a** and the microphone hole **247b** are implemented as one hole, or may include a speaker without the speaker hole **247a** (e.g., a piezo speaker).

(62) According to various embodiments, the camera modules **249a** and **249b** may include a first camera module **249a** (e.g., a front camera) and a second camera module **249b** (e.g., a rear camera) (e.g., a second camera module **249b** in FIGS. 5A and 5B). According to an embodiment, the electronic device **101** may include at least one of a wide-angle camera, a telephoto camera, or a close-up camera, and according to an embodiment, may include an infrared projector and/or an infrared receiver to measure a distance to a subject. The camera modules **249a** and **249b** may include one or multiple lenses, an image sensor, and/or an image signal processor. The first camera module **249a** may be disposed to face the same direction as the display **203**. For example, the first camera module **249a** may be disposed on the periphery of the first display area **A1** or in an area overlapping the display **203**, and when disposed in the area overlapping with the display **203**, may capture an image of a subject through the display **203**. According to an embodiment, the first camera module **249a** may not be visually exposed to a screen display area (e.g., the first display area **A1**), and may include a hidden under-display camera (UDC). According to an embodiment, the second camera module **249b** may capture an image of a subject in a direction opposite to the first display area **A1**. According to an embodiment, the first camera module **249a** and/or the second camera module **249b** may be disposed on the second housing **202**.

(63) According to various embodiments, the indicator (not shown) of the electronic device **101** may be disposed in the first housing **201** or the second housing **202**, and may include a light-emitting diode to provide state information of the electronic device **101** as a visual signal. The sensor module (not shown) of the electronic device **101** may generate an electrical signal or data value corresponding to an internal operation state of the electronic device **101** or an external environmental state. The sensor module may include, for example, a proximity sensor, a fingerprint sensor, or a biometric sensor (e.g., iris/face recognition sensor, or an HRM sensor). In another embodiment, the sensor module may include at least one of, for example, a gesture sensor, a gyro sensor, a barometric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared (IR) sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(64) FIG. 4 is an exploded perspective view of an electronic device according to various embodiments of the disclosure.

(65) FIG. 5A is a cross-sectional view taken along line A-A' in FIG. 2 according to various embodiments of the disclosure.

(66) FIG. 5B is a cross-sectional view taken along line B-B' in FIG. 3 according to various embodiments of the disclosure.

(67) Referring to FIGS. 4, 5A, and/or 5B, the electronic device **101** may include a first housing **201**, a second housing **202**, a display assembly **230**, and a driving structure **240**. The configurations of the first housing **201**, the second housing **202**, and the display assembly **230** in FIGS. 4, 5A, and/or 5B may be wholly or partially identical to the configurations of the first housing **201**, the second housing **202**, and the display **203** in FIGS. 2 and/or 3.

(68) According to various embodiments, the first housing **201** may include a first cover member **211** (e.g., the first cover member **211** in FIGS. 2 and 3), a frame **213**, and a first rear plate **215**.

(69) According to various embodiments, the first cover member **211** may accommodate at least a portion of the frame **213**, and may accommodate a component (e.g., a battery **289**) positioned in the frame **213**. According to one embodiment, the first cover member **211** may be formed to surround at least a portion of the second housing **202**. According to an embodiment, a second circuit board **249** accommodating an electronic component (e.g., the processor **120** and/or the memory **130** in FIG. 1) may be connected to the first cover member **211**.

(70) According to various embodiments, the frame **213** may be connected to the first cover member **211**. For example, the frame **213** may be connected to the first cover member **211**, and the second housing **202** may move relative to the first cover member **211** and/or the frame **213**. According to an embodiment, the frame **213** may accommodate a battery **289** (e.g., the battery **189** in FIG. 1). According to an embodiment, the frame **213** may include a curved portion **213a** facing the display assembly **230**.

(71) According to various embodiments, the first rear plate **215** may substantially form at least a portion of the exterior of the first housing **201** or the electronic device **101**. For example, the first rear plate **215** may be coupled to the outer surface of the first cover member **211**. According to an embodiment, the first rear plate **215** may provide a decorative effect to the exterior of the electronic device **101**. The first rear plate **215** may be manufactured using at least one of metal, glass, synthetic resin, or ceramic.

(72) According to various embodiments, the second housing **202** may include a second cover member **221** (e.g., the second cover member **221** in FIGS. 2 and 3), a rear cover **223**, and a second rear plate **225**.

(73) According to an embodiment, the second cover member **221** may be connected to the first housing **201** through a guide rail **250**, and may reciprocate linearly in one direction (e.g., the direction of arrow {circle around (1)} in FIG. 3) while being guided by the guide rail **250**.

(74) According to various embodiments, the second cover member **221** may support at least a portion of the display **203**. For example, the second cover member **221** may include a first surface **F1**, and a first display area **A1** of the display **203** may be substantially positioned on the first surface **F1** and maintain a flat plate shape. According to an embodiment, the second cover member **221** may be formed of a metal material and/or a non-metal (e.g., polymer) material. According to an embodiment, a first circuit board **248** accommodating an electronic component (e.g., the processor **120** and/or the memory **130** in FIG. 1) may be connected to the second cover member **221**.

(75) According to various embodiments, the rear cover **223** may protect a component (e.g., the first circuit board **248**) positioned on the second cover member **221**. For example, the rear cover **223** may be connected to the second cover member **221** and formed to surround at least a portion of the first circuit board **248**. According to an embodiment, the rear cover **223** may include an antenna pattern for communicating with an external electronic device. For example, the rear cover **223** may include a laser direct structuring (LDS) antenna.

(76) According to various embodiments, the second rear plate **225** may substantially form at least a portion of the exterior of the second housing **202** or the electronic device **101**. For example, the second rear plate **225** may be coupled to the outer surface of the second cover member **221**.

According to one embodiment, the second rear plate **225** may provide a decorative effect to the exterior of the electronic device **101**. The second rear plate **225** may be manufactured using at least one of metal, glass, synthetic resin, or ceramic.

(77) According to various embodiments, the display assembly **230** may include a display **231** (e.g., the display **203** in FIGS. 2 and/or 3) and a multi-bar structure **232** supporting the display **231**. According to an embodiment, the display **231** may be referred to as a flexible display, a foldable display, and/or a rollable display.

(78) According to various embodiments, the multi-bar structure **232** may be connected or attached to at least a portion (e.g., a second display area **A2**) of the display **231**. According to an embodiment, the multi-bar structure **232** may be moved relative to the first housing **201** by sliding of the second housing **202**. When the electronic device **101** is in a closed state (e.g., the state of FIG. 2), the multi-bar structure **232** may be mostly accommodated in the first housing **201**, and may be positioned between the first cover member **211** and the second cover member **221**.

According to an embodiment, at least a portion of the multi-bar structure **232** may move while corresponding to the curved surface **213a** positioned at the edge of the frame **213**. According to an embodiment, the multi-bar structure **232** may be referred to as a display support member or a support structure, and may have the form of one elastic plate.

(79) According to various embodiments, the driving structure **240** may move the second housing **202** relative to the first housing **201**. For example, the driving structure **240** may include a motor **241** configured to generate a driving force for sliding of the housings **201** and **202**. The driving structure **240** may include a gear (e.g., a pinion) connected to the motor **241**, and a rack **242** configured to mesh with the gear.

(80) According to various embodiments, the housing in which the rack **242** is positioned may be different from the housing in which the motor **241** is positioned. According to an embodiment, the motor **241** may be connected to the second housing **202**, and the rack **242** may be connected to the first housing **201**. According to another embodiment, the motor **241** may be connected to the first housing **201**, and the rack **242** may be connected to the second housing **202**.

(81) According to various embodiments, the first housing **201** may accommodate the first circuit board **248** (e.g., the main board). According to an embodiment, a processor (e.g., the processor **120** in FIG. 1), a memory (e.g., the memory **130** in FIG. 1), and/or an interface (e.g., the interface **177** in FIG. 1) may be mounted on the first circuit board **248**. The processor may include, for example, one or more of a central processing unit, an application processor, a graphics processing unit, an image signal processor, a sensor hub processor, or a communication processor. According to various embodiments, the first circuit board **248** may include a flexible printed circuit board type radio frequency cable (FRC). The first circuit board **248** may be disposed on at least a portion of the second cover member **221**, and may be electrically connected to an antenna module (e.g., the antenna module **197** in FIG. 1) and a communication module (e.g., the communication module **190** in FIG. 1).

(82) According to an embodiment, the memory may include, for example, volatile memory or non-volatile memory.

(83) According to an embodiment, the interface may include, for example, a high-definition multimedia interface (HDMI), a universal serial bus (USB) interface, an SD card interface, and/or an audio interface. The interface may, for example, electrically or physically connect the electronic device **101** to an external electronic device, and may include a USB connector, an SD card/MMC connector, or an audio connector.

(84) According to various embodiments, the electronic device **101** may include the second circuit board **249** (e.g., a sub-circuit board) spaced apart from the first circuit board **248** (e.g., a main circuit board) in the first housing **201**. The second circuit board **249** may be electrically connected to the first circuit board **248** through a flexible connection substrate. The second circuit board **249** may be electrically connected to the battery **289** or electrical components disposed in an end area of

the electronic device **101**, such as a speaker and/or a sim socket, to transmit signals and power. According to an embodiment, the second circuit board **249** may accommodate a wireless charging antenna (e.g., a coil). For example, the battery **289** may receive power from an external electronic device by using the wireless charging antenna. In another example, the battery **289** may transmit power to an external electronic device by using the wireless charging antenna.

(85) According to various embodiments, the battery **289** is a device for supplying power to at least one element of the electronic device **101**, and may include a non-rechargeable primary cell, a rechargeable secondary cell, or a fuel cell. The battery **289** may be integrally disposed in the electronic device **101**, and may be detachably disposed in the electronic device **101**. According to an embodiment, the battery **289** may be formed as one integrated battery or may include multiple separate batteries. According to an embodiment, the battery **289** may be positioned in the frame **213**, and the battery **289** may be slid together with the frame **213**.

(86) According to various embodiments, the guide rail **250** may guide the movement of the multi-bar structure **232**. For example, the multi-bar structure **232** may slide along a slit **251** formed in the guide rail **250**. According to an embodiment, the guide rail **250** may be connected to the first housing **201**. For example, the guide rail **250** may be connected to the first cover member **211** and/or the frame **213**. According to an embodiment, the slit **251** may be referred to as a groove or a recess formed in the inner surface of the guide rail **250**.

(87) According to various embodiments, the guide rail **250** may provide pressure to the multi-bar structure **232** based on the driving of the motor **241**.

(88) According to an embodiment, when the electronic device **101** is switched from a closed state to an opened state, the inner portion **252** of the guide rail **250** may provide pressure to the multi-bar structure **232**. The multi-bar structure **232**, provided with pressure, may move along the slit **251** of the guide rail **250**, and the second housing **202** may switch from a slide-in state to a slide-out state with respect to the first housing **201**. At least a portion of the display assembly **230** accommodated between the first cover member **211** and the frame **213** may be extended to the front surface.

(89) According to an embodiment, when the electronic device **101** is switched from an opened state to a closed state, the outer portion **253** of the guide rail **250** may provide pressure to the bent multi-bar structure **232**. The multi-bar structure **232** provided with pressure may move along the slit **251** of the guide rail **250**, and the second housing **202** may switch from a slide-out state to a slide-in state with respect to the first housing **201**. At least a portion of the display assembly **230** may be accommodated between the first cover member **211** and the frame **213**.

(90) Referring to FIG. 5A, when the electronic device **101** is in a closed state, at least a portion of the second housing **202** may be disposed to be accommodated in the first housing **201**. As the second housing **202** is disposed to be accommodated in the first housing **201**, the overall volume of the electronic device **101** may be reduced. According to an embodiment, when the second housing **202** is accommodated in the first housing **201**, the size of the display **231** visually exposed may be reduced. For example, when the second housing **202** is fully accommodated in the first housing **201**, the first display area **A1** of the display **231** may be visually exposed, and the second display area **A2** may not be visually exposed. At least a portion of the second display area **A2** may be disposed between the battery **289** and the rear plates **215** and **225**.

(91) Referring to FIG. 5B, when the electronic device **101** is in an opened state, at least a portion of the second housing **202** may protrude from the first housing **201**. When the second housing **202** protrudes from the first housing **201**, the overall volume of the electronic device **101** may increase. According to an embodiment, when the second housing **202** protrudes from the first housing **201**, at least a portion of the second display area **A2** of the display **231** may be visually exposed to the outside of the electronic device **101** together with the first display area **A1**.

(92) FIG. 6 is an illustrative flowchart illustrating functions or operations in which a property of a one-handed mode screen is changed depending on a grip type in which a user grips an electronic device when the state of an electronic device according to an embodiment herein is changed. In an

embodiment, at least one of the operations described below may be electronically executed.

(93) The embodiment illustrated in FIG. 6 is only one embodiment, and the order of operations according to various embodiments disclosed herein may be different from that illustrated in FIG. 6, and some operations illustrated in FIG. 6 may be omitted. Alternatively, the order thereof may be changed, or operations may be merged.

(94) Referring to FIG. 6, in operation 610, the electronic device 101 according to an embodiment herein may display, on the display 203 (e.g., a flexible display), a one-handed mode screen that is reduced and displayed according to a predetermined ratio corresponding to a first screen (e.g., a home screen or an application execution screen). FIG. 7A is an exemplary view illustrating a screen (e.g., a first screen 710) for configuring a one-handed manipulation mode according to an embodiment herein. Referring to FIG. 7A, the electronic device 101 according to an embodiment herein may configure, through a user input on the screen (e.g., the first screen 710) for configuring a one-handed manipulation mode, whether to use the one-handed manipulation mode and a method for entering the one-handed manipulation mode (e.g., entry via gesture or entry via a specific button). FIG. 7B is an exemplary view illustrating a function or operation of entering a one-handed manipulation mode according to a user gesture according to an embodiment herein. FIG. 7C is an exemplary view illustrating a function or operation of entering a one-handed manipulation mode according to a user's selection input for a visual object according to an embodiment herein.

Referring to FIG. 7B, the electronic device 101 according to an embodiment herein may enter a one-handed manipulation mode when detecting that a gesture 722 (e.g., a downward swipe gesture) that is designated by a user or predesignated is input. As illustrated in FIG. 7B, in response to entering the one-handed manipulation mode, the electronic device 101 according to an embodiment herein may reduce a specific screen, which is currently being displayed on the display 203, to a designated ratio, and may display the reduced screen on the display 203. As illustrated in FIG. 7B, the electronic device 101 according to an embodiment herein may display, on the display 203, the screen (e.g., a one-handed mode screen 730) reduced to the designated ratio to be biased in a specific direction (e.g., right or left). Referring to FIG. 7C, the electronic device 101 according to an embodiment herein may enter the one-handed manipulation mode through a menu related to a virtual keypad 750 of a designated application (e.g., a message application). When a selection input for a designated visual object (e.g., a "one-handed keyboard" icon) displayed on an execution screen (e.g., a second screen 740) of the designated application (e.g., the message application) is detected, the electronic device 101 according to an embodiment herein may display, on the display 203, the virtual keypad 750 having a size reduced according to a designated ratio. As illustrated in FIG. 7C, the electronic device 101 according to an embodiment herein may display, on the display 203, the virtual keypad 750 reduced to the designated ratio to be biased in a specific direction (e.g., right or left). However, according to the art related to various embodiments herein, in the one-handed manipulation mode, the one-handed mode screen (e.g., the reduced execution screen or the reduced virtual keypad) may be displayed at a position at which the same has been previously displayed. In other words, the one-handed mode screen may be displayed regardless of the type of the user's grip. FIGS. 8A and 8B are exemplary views illustrating the art related to embodiments herein, in which a one-hand mode screen is displayed regardless of the type of a user's grip.

Referring to FIG. 8A, a mobile terminal 801 according to the art related to various embodiments herein may display a one-handed mode screen 730 reduced to be biased toward the right that is a position at which the one-handed mode screen has been previously displayed, even though a user grips the mobile terminal 801 with the left hand. In this case, for example, the user of the mobile terminal 801 may experience inconvenience of having to use the one-handed mode screen 730 after changing the position of the one-handed mode screen 730 in a setting menu through a setting icon 732. Alternatively, referring to FIG. 8B, in the mobile terminal 801 according to the art related to the embodiments herein, when the mobile terminal 801 is switched from a closed state to an opened state, the one-handed mode screen 730 is displayed while being extended in the upward

direction even though the user grips the lower portion of the mobile terminal **801**, and thus it may be difficult for the user of the mobile terminal **801** to control an extended one-handed mode screen area **812**. Various embodiments herein may provide the electronic device **101** which may determine the display position or extension direction of the one-handed mode screen **730** according to the type of a user's grip, thereby reducing the user's discomfort and allowing the user to use a one-handed manipulation mode.

(95) In operation **620**, the electronic device **101** according to an embodiment herein may detect that the second housing **202** is moved while the one-handed mode screen **730** is displayed on the display **203**. FIG. **9A** is an exemplary view illustrating a grip sensing area (e.g., at least one among a first grip sensing area **910**, a second grip sensing area **920**, a third grip sensing area **930**, a fourth grip sensing area **940**, a fifth grip sensing area **950**, a sixth grip sensing area **960**, a seventh grip sensing area **970**, and an eighth grip sensing area **980**) according to an embodiment herein.

According to a non-limiting exemplary embodiment, the first grip sensing area **910** may be located at an upper side portion of the second housing **202**, that is, at an upper side portion of the second cover member **221**. The fourth grip sensing area **940** may be located at an upper side portion of the second housing **202**, that is opposed to the upper side portion where the first grip sensing area is located. The second grip sensing area **920** and the third grip sensing area **930** may be located at a central upper portion of the second housing **202**, that is, at a central upper portion of the second cover member **221**. The fifth grip sensing area **950** may be located at a first lateral portion of the first housing **201**, that is, at a first lateral portion of the first cover member **211**. The eighth grip sensing area **980** may be located at a second lateral portion of the first housing **201**, that is opposite to the first lateral portion. The sixth grip sensing area **960** may be located at a lower end portion of the first housing **201**. The seventh grip sensing area **970** may be located at a lower end portion of the first housing **201** opposed to the sixth grip sensing area **960**. Referring to FIG. **9A**, the electronic device **101** according to an embodiment herein may include a grip sensor. The grip sensor according to an embodiment herein may include a sensor for detecting proximity of a portion of a user's body in a capacitive manner. The grip sensor according to an embodiment herein may be provided in an area made of a metal material. The grip sensing area (e.g., at least one among the first grip sensing area **910**, the second grip sensing area **920**, the third grip sensing area **930**, the fourth grip sensing area **940**, the fifth grip sensing area **950**, the sixth grip sensing area **960**, the seventh grip sensing area **970**, and the eighth grip sensing area **980**) may include an area having a grip sensor. The grip sensing area (e.g., at least one among the first grip sensing area **910**, the second grip sensing area **920**, the third grip sensing area **930**, the fourth grip sensing area **940**, the fifth grip sensing area **950**, the sixth grip sensing area **960**, the seventh grip sensing area **970**, and the eighth grip sensing area **980**) according to an embodiment herein may be distinguished by, for example, metal segments **905**. The grip sensor according to an embodiment herein may not be provided on the metal segments **905**. As exemplarily illustrated in FIG. **9A**, the grip sensor according to an embodiment herein may be provided on lateral metal portions of the first cover member **211** and the second cover member **221**. FIGS. **9B** and **9C** are exemplary views illustrating a grip sensing area (e.g., at least one among a first grip sensing area **910**, a second grip sensing area **920**, a third grip sensing area **930**, a fourth grip sensing area **940**, a fifth grip sensing area **950**, a sixth grip sensing area **960**, a seventh grip sensing area **970**, and an eighth grip sensing area **980**) of an electronic device according to another embodiment herein. Referring to FIGS. **9B** and **9C**, a first cover member **211** of an electronic device **101** according to another embodiment herein may be formed to contact a second cover member **221** while covering a portion of the second cover member **221** without enclosing all of the second cover member **221**. As exemplarily illustrated in FIG. **9C**, the electronic device **101** according to another embodiment herein may include a grip sensing area (e.g., at least one among the first grip sensing area **910**, the second grip sensing area **920**, the third grip sensing area **930**, the fourth grip sensing area **940**, the fifth grip sensing area **950**, the sixth grip sensing area **960**, the seventh grip sensing area **970**, and the eighth grip sensing

area **980**).

(96) In operation **630**, the electronic device **101** according to an embodiment herein may determine, based on the detection of the movement of the second housing **202**, the direction of extension of the one-handed mode screen **730** according to whether a portion of the user's body grips the upper end (e.g., the first grip sensing area **910**) of the second housing **202**. In operation **640**, the electronic device **101** according to an embodiment herein may extend and display the one-handed mode screen **730** according to the extension direction determined in operation **630**. FIG. **10A** is an exemplary view illustrating a function or operation in which the extension direction of the one-handed mode screen **730** is determined based on a first grip type when the electronic device **101** according to an embodiment herein is switched from a closed state to an opened state. Referring to FIG. **10A**, the user of the electronic device **101** according to an embodiment herein may not grip the upper end (e.g., an area including at least a portion of the first grip sensing area **910**) of the electronic device **101**. Herein, for convenience of description, the state in which the user does not grip the upper end (e.g., the area including at least a portion of the first grip sensing area **910**) of the electronic device **101** may be referred to as a "first grip type". When the user grips the electronic device **101** in the first grip type, the electronic device **101** according to an embodiment herein may display the one-handed mode screen **730** on the display **203** while extending the one-handed mode screen **730** downward as illustrated in FIG. **10A**. The extended one-handed mode screen **730** according to an embodiment herein may include a screen newly displayed on the display **203** according to the expansion of the display **203**. The first grip type according to an embodiment herein may include, for example, a grip type in which the user's grip is detected in an area including the fifth grip sensing area **950** and the sixth grip sensing area **960**, but the user's grip is not detected in the first grip sensing area **910**. In the first grip type according to an embodiment herein, the extension direction of the electronic device **101** and the extension direction of the one-handed mode screen **730** may be opposite to each other.

(97) FIG. **10B** is an exemplary view illustrating a function or operation in which the extension direction of a one-handed mode screen is determined based on a second grip type when an electronic device according to an embodiment herein is switched from a closed state to an opened state. Referring to FIG. **10B**, a user of the electronic device **101** according to an embodiment herein may be gripping the upper end (e.g., an area including at least a portion of the first grip sensing area **910**) of the electronic device **101**. Herein, for convenience of description, the state in which the user grips the upper end (e.g., the area including at least a portion of the first grip sensing area **910**) of the electronic device **101** may be referred to as a "second grip type". When the user grips the electronic device **101** in the second grip type, the electronic device **101** according to an embodiment herein may display the one-handed mode screen **730** on the display **203** while extending the one-handed mode screen **730** upward as illustrated in FIG. **10B**. The extended one-handed mode screen **730** according to an embodiment herein may include a screen newly displayed on the display **203** according to the extension of the display **203**. The second grip type according to an embodiment herein may include, for example, a grip type in which the user's grip is detected in an area including the first grip sensing area **910**, the fifth grip sensing area **950**, and the sixth grip sensing area **960**. In the second grip type according to an embodiment herein, the extension direction of the electronic device **101** and the extension direction of the one-handed mode screen **730** may be identical to each other. The display position of the one-handed mode screen **730** (e.g., the upper or lower end of the one-handed mode screen **730**) according to an embodiment herein may be maintained while the one-handed mode screen **730** is extended. In other words, the display position of the one-handed mode screen **730** may be moved by the extension of the display **203**, and thus it may appear to the user that the display position of the one-handed mode screen **730** is maintained despite the extension of the display **203**.

(98) FIG. **11A** is an exemplary view illustrating a function or operation in which the extension direction of a one-handed mode screen **730** is determined based on a second grip type when an

electronic device **101** according to an embodiment herein is switched from an opened state to a closed state. Referring to FIG. **11A**, when the type of a user's grip on the electronic device **101** is a second grip type and when the electronic device **101** is switched from the opened state to the closed state, the electronic device **101** according to an embodiment herein may display the one-handed mode screen **730** on the display **203** while reducing the one-handed mode screen **730** in a direction opposite to the driving direction of the electronic device **101**. FIG. **11B** is an exemplary view illustrating a function or operation in which the extension direction of the one-handed mode screen **730** is determined based on a first grip type when the electronic device **101** according to an embodiment herein is switched from an opened state to a closed state. Referring to FIG. **11B**, when the type of a user's grip on the electronic device **101** is a first grip type and when the electronic device **101** is switched from the opened state to the closed state, the electronic device **101** according to an embodiment herein may display the one-handed mode screen **730** on the display **203** while reducing the one-handed mode screen **730** in a direction (e.g., a downward direction) identical to the driving direction of the electronic device **101**. The display position of the one-handed mode screen **730** (e.g., the upper or lower end of the one-handed mode screen **730**) according to an embodiment herein may be maintained while the one-handed mode screen **730** is reduced. In other words, the display position of the one-handed mode screen **730** may be moved by the reduction of the display **203**, and thus it may appear to the user that the display position of the one-handed mode screen **730** is maintained despite the reduction of the display **203**.

(99) FIG. **12** is an exemplary flowchart illustrating a function or operation in which a one-handed mode screen **730** is moved and displayed according to the type of a user's grip when the electronic device **101** according to an embodiment herein is in a closed state. FIGS. **13A** to **13C** are exemplary views illustrating the function or operation exemplarily described in FIG. **12** in terms of a user interface.

(100) In an embodiment, at least one of the operations to be described below may be executed by the processor **120** of the electronic device **101**.

(101) The embodiment illustrated in FIG. **12** is only one embodiment, and the order of operations according to various embodiments disclosed herein may be different from that illustrated in FIG. **12**. Some operations illustrated in FIG. **12** may be omitted, the order of the operations may be changed, or the operations may be merged.

(102) Referring to FIG. **12**, in operation **1210**, the electronic device **101** according to an embodiment herein may display, on the display **203** (e.g., a flexible display), a one-handed mode screen that is reduced and displayed according to a designated ratio corresponding to a first screen (e.g., a home screen or an application execution screen). As illustrated in FIG. **13A**, the electronic device **101** according to an embodiment herein may display the one-handed mode screen **730** on the display **203**.

(103) In operation **1220**, the electronic device **101** according to an embodiment herein may determine the type of a user's grip. The electronic device **101** according to an embodiment herein may determine, based on a value sensed by a grip sensor, whether the type of the user's grip is a first grip type or a second grip type.

(104) In operation **1230**, the electronic device **101** according to an embodiment herein may move and display the one-handed mode screen **730**, based on the grip type determined in operation **1220**. Referring to FIG. **13A**, when it is determined that the type of the user's grip on the electronic device **101** is the first grip type, the electronic device **101** according to an embodiment herein may move a display position of the one-handed mode screen **730** from a first position (e.g., a position where the one-handed mode screen **730** was last displayed at the end of the previous one-handed mode) to a designated second position (e.g., a position below the first position). Alternatively, referring to FIG. **13B**, when it is determined that the type of the user's grip on the electronic device **101** is the second grip type, the electronic device **101** according to an embodiment herein may move the display position of the one-handed mode screen **730** from a third position (e.g., a position

where the one-handed mode screen **730** was last displayed at the end of the previous one-handed mode) to a designated fourth position (e.g., a position above the third position). Referring to FIG. **13C**, when it is determined that the type of the user's grip on the electronic device **101** is the first grip type, the electronic device **101** according to an embodiment herein may move the display position of the one-handed mode screen **730** from a fifth position (e.g., a position where the one-handed mode screen **730** was last displayed at the end of the previous one-handed mode) to a designated sixth position (e.g., a position to the left of the first position) depending on the type of hand (e.g., the left hand) gripping the electronic device **101**. The electronic device **101** according to an embodiment herein may determine whether a user's hand gripping the electronic device **101** is a left hand or a right hand, by using a pattern of sensing data sensed by the grip sensor. For example, the electronic device **101** according to an embodiment herein may compare the similarities between the pattern of predetermined sensing data and the pattern of the sensing data sensed by the grip sensor to determine whether the user's hand gripping the electronic device **101** is a left hand or a right hand.

(105) According to another embodiment herein, operation **1220** may be performed before operation **1210**. For example, the electronic device **101** according to an embodiment herein may determine a display position of the one-handed mode screen **730** after determining the type of the user's grip, and may then display the one-handed mode screen **730** in the determined display position.

(106) An electronic device (e.g., the electronic device **101** in FIG. **1**) according to an embodiment herein may include a first housing (e.g., the first housing **201** in FIG. **2A**), a second housing (e.g., the second housing **202** in FIG. **2**) disposed to be movable relative to the first housing, at least one processor (e.g., the processor **120** in FIG. **1**) disposed in the first housing or the second housing, grip sensors respectively disposed on edges of the first housing and the second housing, and a flexible display (e.g., the display **160** in FIG. **1**) at least partially mounted to the surface of the second housing and at least partially exposed to the outside of the electronic device, wherein a portion of the flexible display is inserted into or withdrawn from the first housing depending on sliding of the second housing, and at least one processor is configured to display, on the flexible display, a one-handed mode screen (e.g., the one-handed mode screen **730** in FIG. **10B**) that is reduced and displayed according to a designated ratio corresponding to a first screen, detect that the second housing is moved while the one-handed mode screen is displayed, and determine, based on the detection of the movement of the second housing, an extension direction of the one-handed mode screen according to whether a first grip sensor, disposed at the edge of the first housing, among the grip sensors senses that a portion of the user's body grips the upper end of the first housing.

(107) According to embodiment of the disclosure, the at least one processor may be further configured to extend and display the one-handed mode screen in a direction corresponding to an extension direction of the first housing when an upper end of the first housing is determined to be gripped.

(108) The at least one processor may be further configured to extend and display the one-handed mode screen in a direction opposite to the extension direction of the first housing when the upper end of the first housing is determined not to be gripped.

(109) The at least one processor may be further configured to display the one-handed mode screen on a first designated area of the flexible display when the upper end of the first housing is determined, through a first grip sensor, to be gripped while the electronic device is in a closed state.

(110) The at least one processor may be further configured to display the one-handed mode screen on a second designated area of the flexible display when the upper end of the first housing is determined, through the first grip sensor, not to be gripped while the electronic device is in the closed state.

(111) The first area may be positioned above the second area.

(112) The at least one processor may be further configured to move and display a display position of

the one-handed mode screen from a currently displayed position to a position close to the upper end when a grip on the upper end is determined to have been sensed while the electronic device is in the closed state.

(113) The at least one processor may be further configured to move and display a display position of the one-handed mode screen from a currently displayed position to a position close to a lower end opposite to the upper end when a grip on the upper end is determined not to have been sensed while the electronic device is in the closed state.

(114) The at least one processor may be further configured to move and display the one-handed mode screen from a currently displayed position to a specified position depending on whether the user's hand gripping the electronic device is a left hand or a right hand when the type of grip on the electronic device is identified as a first grip type while the electronic device is in the closed state.

(115) The one-handed mode screen may comprise a screen having a size smaller than a size of at least one virtual keypad or the flexible display exposed through the electronic device.

(116) According to embodiment of the disclosure, the method may further comprise extending and displaying the one-handed mode screen in a direction corresponding to an extension direction of the first housing when an upper end of the first housing is determined to be gripped.

(117) The method may further comprise extending and displaying the one-handed mode screen in a direction opposite to the extension direction of the first housing when the upper end of the first housing is determined not to be gripped.

(118) The method may further comprise displaying the one-handed mode screen on a first designated area of the flexible display when the upper end of the first housing is determined, through a first grip sensor, to be gripped while the electronic device is in a closed state.

(119) The method may further comprise displaying the one-handed mode screen on a second designated area of the flexible display when the upper end of the first housing is determined, through the first grip sensor, not to be gripped while the electronic device is in the closed state.

(120) The first area may be positioned above the second area.

(121) The method may further comprise moving and displaying a display position of the one-handed mode screen from a currently displayed position to a position close to the upper end when a grip on the upper end is determined to have been sensed while the electronic device is in the closed state.

(122) The method may further comprise moving and displaying a display position of the one-handed mode screen from a currently displayed position to a position close to a lower end opposite to the upper end when a grip on the upper end is determined not to have been sensed while the electronic device is in the closed state.

(123) The method may further comprise moving and displaying the one-handed mode screen from a currently displayed position to a specified position depending on whether the user's hand gripping the electronic device is a left hand or a right hand when the type of grip on the electronic device is identified as a first grip type while the electronic device is in the closed state.

(124) The one-handed mode screen may comprise a screen having a size smaller than a size of at least one virtual keypad or the flexible display exposed through the electronic device.

(125) The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

(126) It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun

corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B”, “at least one of A and B”, “at least one of A or B”, “A, B, or C”, “at least one of A, B, and C”, and “at least one of A, B, or C”, may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd”, or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with”, “coupled to”, “connected with”, or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

(127) As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic”, “logic block”, “part”, or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(128) Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

(129) According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

(130) According to an embodiment of the disclosure, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to an embodiment of the disclosure, one or more of the above-described components or operations may be omitted, or one or more other components or operations may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to an embodiment of the disclosure, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

Claims

1. An electronic device comprising: a first housing; a second housing disposed to be movable with respect to the first housing; at least one grip sensor positioned on at least one edge of both the first housing and the second housing; a flexible display at least partially mounted on a surface of the second housing and adapted to be inserted into or withdrawn from the first housing through slide-in or slide-out movement of the second housing; and at least one processor operably connected to the first housing, the second housing, the at least one grip sensor, and the flexible display, memory storing instructions that, when executed by the at least one processor, cause the electronic device to: display, on the flexible display, a one-handed mode screen that is reduced in size according to a specified ratio corresponding to the full screen in a slide-in state, detect that the second housing is moved while the one-handed mode screen is displayed, detect a grip state of a user holding the electronic device, and move a display position of the one-handed mode screen based on the detected grip state of the user.
2. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor, further cause the electronic device to extend and display the one-handed mode screen in a direction corresponding to an extension direction of the first housing when an upper end of the first housing is determined to be gripped.
3. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor, further cause the electronic device to extend and display the one-handed mode screen in a direction opposite to an extension direction of the first housing when an upper end of the first housing is determined not to be gripped.
4. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor, further cause the electronic device to display the one-handed mode screen on a first designated area of the flexible display when an upper end of the first housing is determined, through a first grip sensor, to be gripped while the electronic device is in a closed state.
5. The electronic device of claim 4, wherein the instructions, when executed by the at least one processor, further cause the electronic device to display the one-handed mode screen on a second designated area of the flexible display when the upper end of the first housing is determined, through the first grip sensor, not to be gripped while the electronic device is in the closed state.
6. The electronic device of claim 5, wherein the first designated area is positioned above the second designated area.
7. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor, further cause the electronic device to move and display a display position of the one-handed mode screen from a currently displayed position to a position close to an upper end when a grip on the upper end is determined to have been sensed while the electronic device is in a closed state.
8. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor, further cause the electronic device to move and display a display position of the one-handed mode screen from a currently displayed position to a position close to a lower end opposite to an upper end when a grip on the upper end is determined not to have been sensed while the electronic device is in a closed state.
9. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor, further cause the electronic device to move and display the one-handed mode screen from a currently displayed position to a specified position depending on whether a user's hand gripping the electronic device is a left hand or a right hand when a type of grip on the electronic device is identified as a first grip type while the electronic device is in a closed state.
10. The electronic device of claim 1, wherein the one-handed mode screen comprises a screen having a size smaller than a size of at least one virtual keypad or the flexible display exposed

through the electronic device.

11. A method for controlling an electronic device, the method comprising: displaying, on a flexible display, a one-handed mode screen that is reduced in size according to a predetermined ratio corresponding to the full screen in a slide-in state; detecting that a second housing is moved while the one-handed mode screen is displayed; detecting a grip state of a user holding on the electronic device; and moving a display position of the one-handed mode screen based on the detected grip state of the user.

12. The method of claim 11, further comprising extending and displaying the one-handed mode screen in a direction corresponding to an extension direction of a first housing when an upper end of the first housing is determined to be gripped.

13. The method of claim 11, further comprising extending and displaying the one-handed mode screen in a direction opposite to an extension direction of a first housing when an upper end of the first housing is determined not to be gripped.

14. The method of claim 11, further comprising displaying the one-handed mode screen on a first designated area of the flexible display when an upper end of a first housing is determined, through a first grip sensor, to be gripped while the electronic device is in a closed state.

15. The method of claim 14, further comprising displaying the one-handed mode screen on a second designated area of the flexible display when the upper end of the first housing is determined, through the first grip sensor, not to be gripped while the electronic device is in the closed state.
